Calculus done right

first edition

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Preface to the first edition

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I

Precalculus

Chapter 1

Functions

Section 1.1

Function basics

Before we start on the calculus topics, this part of the book will acquaint you with the mathematics you would need to understand in order to read this book. If you are familiar with the these topics, you may skip to the Part II on Calculus I. For the rest of you, the pace of this part will be very fast, it is recommended that you read up on concepts you do understand very well. We will start with the idea of a *function*. By now, I am sure that many of you may have seen equations like these:

$$x^2 - 5x + 6 = 0 \tag{1.1}$$

$$\sin(x) = 0.5, 0 \leq x \leq \frac{\pi}{2} \tag{1.2}$$

$$y = mx + c \tag{1.3}$$

Out of the three equations, notice that Equation (1.3) is special. There is not some fixed x values that you must plug in. Instead, the y value varies with the x value. Given a x value, the equation "produces" a y value. This brings us to the idea of a function. A function is simply something that takes in a value, and "produces" another value based on the input. You may have came across functions before, such as the trigonometric functions like sin and cos. The value of $\sin(x)$ varies with different values of x. Of course, it is time for me to give you the definition of a function.

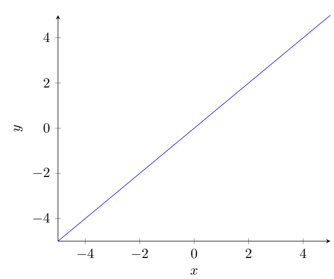


Figure 1.1. Graph of y=x.

Definition 1.1 A function f is a mathematical object that assigns its inputs, or *domain*, to its outputs, or *range*. The notation $f : A \to B$ means that the function has a domain A and range B. The notation $f(a)$ where $a \in A$ denotes the element of B to which a is assigned to.

Notice that A and B are sets.

Example Here are some examples of functions:

$$f(x) = x + 1 \tag{1.4}$$

$$g(x) = \sin(x) \tag{1.5}$$

$$h(x) = x^2 - 5x + 6 \tag{1.6}$$

To evaluate the output of a function, you simply substitute the input to the expression that defines the function. Notice that for Equation (1.4), it is simply the line $y = x + 1$. Each value of $f(x)$ for some $x \in \mathbb{R}$ is just y. Take $x = 1$ for example, $f(1) = 1 + 1 = 2$ and for the line $y = x + 1$, notice the y value here is also 2. The function assigns each x value to $x + 1$. Hence, in this case, $f : \mathbb{R} \to \mathbb{R}$. For Equation (1.5) , the function takes any real x value and maps it to $\sin(x)$. As a result, its domain would be $\mathbb{R}$ and its range would be $[-1, 1]$. Again, it is just like the graph

$y = \sin(x)$.

Lastly, the quadratic in Equation (1.6) is another real function(i.e. it has a real domain and range). Think about what happens when $h(x) = 0$, what value of x must you plug in to make $h(x) = 0$. Well, isn't it just the roots of the quadratic. Hence, the roots of a function are just the values x such that $h(x) = 0$. In this case, it would be 2 and 3. Thus, $h(2) = h(3) = 0$.

Actually, functions can be more than the usual $f(x) =$ something. Let's say you want to have a function that outputs "1" if the input is rational and "0" if it is not, you can define it as such:

This is actually the famous Dirichlet function that shatters certain intuitions on functions and will be discussed in later chapters.

$$f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \\ 0, & \text{if } x \notin \mathbb{Q} \end{cases} \tag{1.7}$$

Here, $f(\frac{1}{2})$ would be 1 but $f(\sqrt{2})$ would be 0. This is an example of having conditionals in a function. Lastly, functions may take more than one input and produce more than one output, examples include:

$$f(x, y) = x^2 + y^2 \tag{1.8}$$

$$g(t) = (\cos(t), \sin(t)) \tag{1.9}$$

Equation (1.8) is an example of a *multivariate* function, in other words, it takes in multiple inputs. If $x = 2$ and $y = 2$, then $f(2, 2) = 8$. On the other hand, Equation (1.9) does not have a single output. Instead, it has a tuple as its output. Evaluating it at $t = \pi$, notice that the output is $(-1, 0)$. What is interesting is that if you trace out its outputs on a plane for all $t \in \mathbb{R}$, you will get:

As you will see in Part IV of the book, this is the parametric equation of a circle.

A tuple is basically a set where order matters. You can treat it as if it is a coordinate.

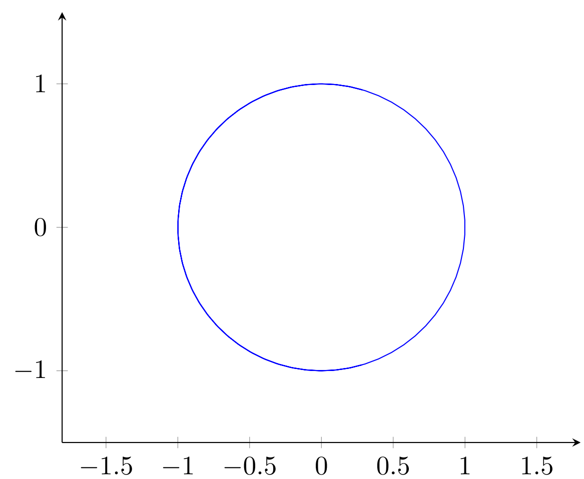


Figure 1.2. Plot of Equation (1.9) for all $t \in \mathbb{R}$

Functions are the building blocks of mathematics, and this book will deal extensively with it. Therefore, it is paramount that you get the basics right. Try the following exercise:

Exercise

1. Given that $x = 2$ and $y = \pi$, evaluate $f(x)$ for:
   (a) $f(x) = x^3 + 1$
   (b) $f(x) = 2^x$

(c) $f(x) = \begin{cases} x^4 + 1 & \textit{if } x < 2 \\ 3^x & \textit{otherwise} \end{cases}$

(d) $f(x, y) = \frac{\sin(y)}{x}$

2. Given the following functions, write down $f : A \to B$ where A is the domain and B is the range(do not consider complex inputs and outputs):

   (a) $f(x) = x$

   (b) $f(x) = \frac{1}{x}$

   (c) $f(x) = \sqrt{x}$

*This function has a special name, the **identity** function. Notice how its input and output are the same.*

3. Find the root(s) of the following functions:

   (a) $f(x) = x^2$

   (b) $f(x) = 2^x - 1$

   (c) $f(x) = \sin(x)$

Solution

1. (a) 9
   (b) 4
   (c) 9
   (d) 0

2. (a) $f : \mathbb{R} \to \mathbb{R}$
   (b) $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ (Note the function is not defined at $x = 0$)
   (c) $f : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ (Square root not defined for negative numbers)

3. (a) 0
   (b) 1
   (c) $n\pi$, $\forall n \in \mathbb{Z}$

SECTION 1.2

Function composition and inverses

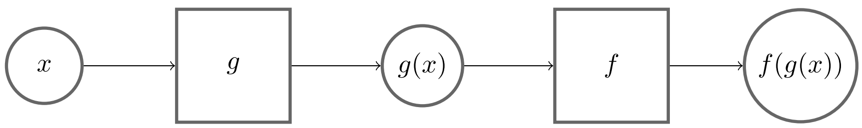
In this chapter, we will look at composition of functions and inverse functions. You can use the normal arithmetic operations on functions like how you would to numbers:

Example If $x = 2$ and $f(x) = x^2 + 4$:

$$f(x) + f(x) = 2^2 + 4 + 2^2 + 4 = 16$$

$$\frac{f(x)}{2} = \frac{2^2 + 4}{2} = 4$$

What is more interesting, perhaps, is taking the *output* of a function and use it as the *input* to another function. You may also think of it as "piping" one function into another. This is called *function composition*.



Note that the rectangles are functions and circles are values.

Figure 1.3. Visual representation of function composition.

Another notation for composing functions together(which we will not really use) is as such:

$$f(g(x)) = (f \circ g)(x)$$

When you compose the same function n-times, the notation for it is:

$$f^n(x) = \underbrace{f(f(\ldots f(x)))}_{n-times}$$

However, this does not apply to trigonometric functions. $\sin^2(x)$ *is* $\sin(x) \times \sin(x)$ *instead of* $\sin(\sin(x))$.

Example Here are some examples of composing functions together: If $x = 3$ and $f(x) = \pi x$, $g(x) = \sin(x) + 1$:

$$g(f(x)) = \sin(\pi \times 3) + 1 = 1 \tag{1.10}$$

$$f^3(x) = \pi^3 x \tag{1.11}$$

There is nothing much here, really, just evaluate the function from the innermost function to the outermost.

Theorem 1.1 Given functions $f : X \to Y$, $g : Y \to Z$ and $h : Z \to W$, prove that the composition of the three functions is *associative*, in other words:

$$h \circ (g \circ f(x)) = (h \circ g) \circ f(x)$$

PROOF Let $x \in X$, we have:

$$h \circ (g \circ f(x)) = h(g \circ f(x)) = h(g(f(x))) = h \circ g(f(x)) = (h \circ g)f(x)$$

This proves the associativity of function composition. QED

Remark This is just a fun little exercise for the reader and ultimately is not that important to the entire book, just a good identity to keep in the back of your mind.

Next, we will delve into inverse functions. Let us start with one I believe you may have seen before, look at the following figure:

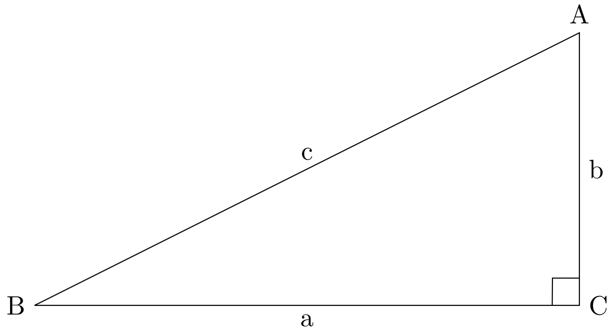


Figure 1.4. Figure of right triangle ABC.

If I ask you what is $\arcsin(B)$, you may say it is $\frac{b}{c}$. But, what is $\arcsin(\sin(B))$, well, that is just B, isn't it. Hence, the arcsin sort of "cancels" the sin. We call arcsin the *inverse function* or just inverse of sin. Generally, given a function $f : X \to Y$, f^{-1} denotes the inverse function that satisfies:

We will use arcsin *as the inverse of* sin *instead of* $\sin^{-1}$ *as it is clearer.*

$$f \circ f^{-1}(y) = y, y \in Y \text{ and } f^{-1} \circ f(x) = x, x \in X \tag{1.12}$$

To find the inverse function of a function f, we have to find the function that maps the outputs of the f back to its inputs. To solve algebraically, we set $f(x)$ for some arbitrary x to y, and make x the subject of the equation.

Example

Given that $f(x) = \frac{x+7}{3}$:

$$\begin{aligned} f(x) &= y \\ \frac{x+7}{3} &= y \\ x+7 &= 3y \\ x &= 3y-7 \end{aligned}$$

Hence, the inverse function is $f^{-1}(y) = 3y - 7$.

Exercise

1. Given $f(x) = \frac{x^2+3}{2}$ and $g(x) = \tan(x) + 1$, evaluate:
   (a) $f(g(\frac{\pi}{4}))$
   (b) $f^3(1)$
   (c) $f(g(\arcsin(0)))$
2. Given that $h(x) = (x-7)^2$, $x \geq 7$, find the inverse of h.
3. Hence, evaluate:
   (a) $h^{-1}(1)$
   (b) $h(h^{-1}(69))$

Solution

1. (a) $\frac{7}{2}$
   (b) $\frac{61}{8}$
   (c) 2
2. $h^{-1}(y) = \sqrt{y} + 7$

3. (a) 8
   (b) 69 (Inverse function and the original function "cancels".)

SECTION 1.3
Injection, surjection and bijection

You may have noticed in the previous exercise that question 2 is particularly interesting. There is a restriction on the domain of $h(x)$. Why is that? This is what this chapter is all about. However, before we explain that, I will go over some definitions of properties of functions. Firstly, we have *injection.*

Definition 1.2 An injective function, f, or a *one-to-one* function, is a function such that every distinct output of the function corresponds to only *one* distinct input, that is:

$$f(x_1) = f(x_2) \text{ implies } x_1 = x_2$$

By contraposition, $x_1 \neq x_2$ *implies* $f(x_1) \neq f(x_2)$.

Of course, this explanation might not have been very clear. Here is a visualisation:

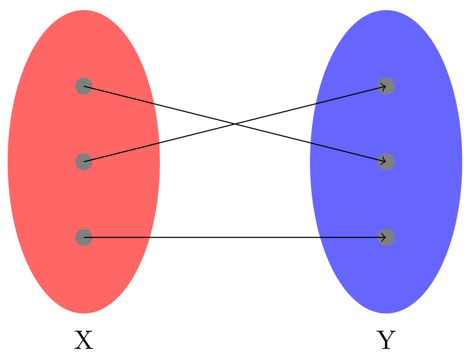


Figure 1.5. Illustration of an injective function with domain X and range Y.

As you can see in Figure 1.5, every output in Y of the function is associated with only one input in X, hence the function is *injective*. To prove that a function f is injective, you can either show that if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$, or if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

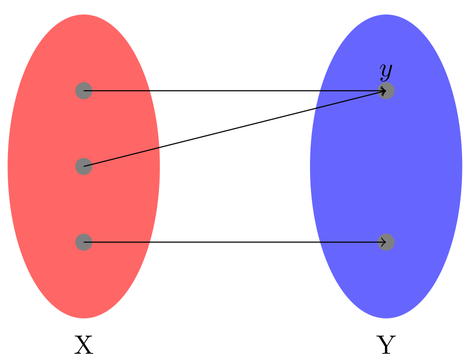


Figure 1.6. Illustration of a non-injective function with domain X and range Y.

Here, the function is *not* injective. As you can see in Figure 1.6, the output y is associated with more than one input in X. With this in mind, it is time to talk about *surjection.*

Definition 1.3 A surjective function f is a function such that for every output y, there exists at least one input x such that $f(x) = y$.

From Figure 1.5 and Figure 1.6, both functions are surjective as every output is associated with at least one input. Do take note that it does not matter how many inputs associate to one output, as long every output is associated with at least one input, the function is surjective. To prove that a function is surjective, you must show that there exists an element in the domain of the function for any arbitrary output in the range.

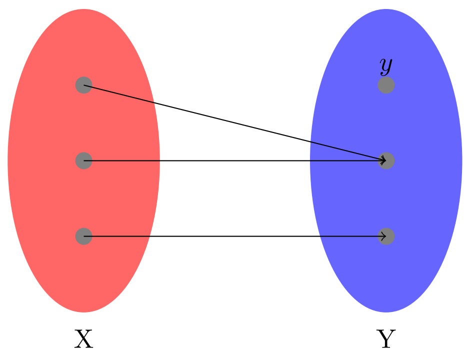


Figure 1.7. Illustration of non-surjective function with domain X and range Y.

From Figure 1.7, we can see that y has no input associated with it. As a result, the function here is not surjective. Lastly, *bijective* functions can't be any easier.

Definition 1.4 A bijective function is a function that is both injective and surjective.

However, bijective functions have a superpower. A function is only bijective *if and only if* it has an inverse.

P iff Q, or "P if and only if Q", means that "if P then Q" and "if Q then P".

Theorem 1.2 A function, $f : A \to B$, has an inverse if and only if it is bijection.

PROOF To begin such proofs where there is a "if and only if" clause, we will prove both cases separately. First, we claim that if the function has an inverse, then it is bijective. To prove this, we simply prove that f is injective and surjective (See Lemma 1.1). Then, we claim that if a function is bijective, then it has an inverse (See Lemma 1.2). With both proofs, we show that f has an inverse if and only if it is a bijection. QED

Lemma 1.1 If a function, $f : A \to B$, has an inverse, then it is bijective.

PROOF Suppose f^{-1} is the inverse of f. Notice that $f^{-1}(f(x))$ is injective as it is simply $f(x) = x$. Suppose $a, a' \in A$ and $f(a) = f(a')$, then $f^{-1}(f(a)) = f^{-1}(f(a')))$. Notice that by injectivity of $f^{-1}(f(x))$, $a = a'$. Hence, f is injective. Then, suppose $b \in B$, we wish to show that the function is surjective. Also, $f(f^{-1}(x)$ is surjective(it is the identity function). Let $f^{-1}(b) = x$ and $f(f^{-1}(b)) = c$ for some $c \in B$. By a simple substitution, $f(f^{-1}(b)) = f(x) = c$, thereby showing there exists a $x \in A$ for every $c \in B$. The two cases proves that f is bijective if it has an inverse. QED

Lemma 1.2 If a function, $f : A \to B$, is a bijection, then it has an inverse.

PROOF Let us define f^{-1} as $f^{-1} : B \to A$. Let $b \in B$. Since f is surjective, there exists $a \in A$ such that $f(a) = b$. Let $f^{-1}(b) = a$. Since f is injective, f^{-1} is well-defined, or a function that outputs the same value given the same input. Next, we will show that f^{-1} is indeed the inverse of f by showing the composition of the two is an identity function. Again, let $a \in A$ and $b = f(a)$ which implies, by definition, $f^{-1}(b) = a$. Composing the two functions, $f(f^{-1}(b)) = f(a) = b$, which shows that f^{-1} is indeed the inverse function. The other case of $f^{-1} \circ f$ is similar. Hence, if f is a bijection, then it has an inverse. QED